Ayush Suryawanshi

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PROFESSIONAL SUMMARY

Chemical engineer with an M.Tech in Catalysis Technology from IIT Madras, working across catalysis, electrochemistry and process engineering. Synthesised Bi-modified Pt/C catalysts, including in the KinCat group at NTNU, prepared inks and electrodes, and tested them in three-electrode cells on an Autolab potentiostat (CV, chronoamperometry, ECSA, durability). Modelled SNCR kinetics in Python (Cantera) with Aspen HYSYS process simulation, completed a DFT course project, and at LOXIM Industries led a two-month study that proposed a lower-cost effluent treatment train. Careful about safety, evidence and the limits of results.

EDUCATION

Master of Technology, Catalysis Technology — CGPA 8.51/10 | Jun 2024 – May 2026

Indian Institute of Technology Madras, Chennai

Coursework: Advanced Chemical Engineering Thermodynamics, Chemical Reactor Theory, Transport Phenomena, Experimental Methods in Catalysis, Catalyst Preparation and Characterization Lab, Computational Methods in Catalysis.

Semester Exchange, Chemical Engineering | Aug 2025 – Dec 2025

Norwegian University of Science and Technology (NTNU), Trondheim, Norway; UTFORSK mobility scholarship

Bachelor of Technology, Chemical Engineering — CGPA 8.17/10 | 2019 – 2023

Indore Institute of Science and Technology, Indore, affiliated to RGPV Bhopal; First Division with Honours

Coursework: Material and Energy Balance, Chemical Reaction Engineering, Heat and Mass Transfer, Fluid Particle Mechanics, Process Equipment Design I and II, Chemical Process Modeling and Simulation, Process Safety and Hazards Management, Process Plant Economics and Management.

Senior School Certificate, Class XII, CBSE — 84.2% | 2019

Delhi Public School, Chhindwara

PROFESSIONAL EXPERIENCE

Management Trainee | Jul 2026 – Present

LOXIM Industries Ltd., Ahmedabad

- Led a two-month study on treating a high-sulphate dye effluent: built a stage-wise cost baseline and compared electrochemical, membrane and crystallisation routes.
- Evaluated electrocoagulation and electroflocculation as pretreatment options; ran screening, sample trials and feasibility checks with more than ten technology vendors.
- Proposed a treatment train that recovers the waste salt as a product (DWSIM simulation, stage-wise OPEX model), at about two-thirds lower modelled operating cost; handed over with a validation roadmap.
- Trained on HPLC, GC-MS, FTIR and UV-Vis; completed plant safety induction and HSE training.

RESEARCH EXPERIENCE

M.Tech Thesis: Bi-Modified Pt/C Anode Catalysts for Formic Acid Fuel Cells | May 2025 – May 2026

IIT Madras, Chennai | Guide: Prof. Raghuram Chetty

- Synthesised a Bi-modified Pt/C series from commercial Pt/C at four Bi precursor concentrations (1.25–10 mM) by aqueous adsorption, NaBH4 reduction and heat treatment at 300 °C under argon.
- Assembled and ran three-electrode cells (glassy carbon and carbon paper working electrodes, Pt counter, Ag/AgCl reference) on an Autolab potentiostat for CV, chronoamperometry, ECSA and mass activity.
- Optimised ionomer loading: cutting Nafion in the catalyst ink from 20 µL to 5 µL raised the formic acid oxidation peak current by about 2.5–2.8 times.
- In one-hour holds at 0.55 V vs RHE, Bi(1.25)-Pt/C retained **92.9%** of its current versus 58.5% for commercial Pt/C; with 1 mM AQSA mediator, retention fell for every catalyst (70.4% and 55.6%).
- Identified an activity-durability trade-off: Bi(5)-Pt/C gave the highest peak mass activity but retained only 52.8%; related Bi loading to morphology and structure by FE-SEM, EDX, XRD and BET.

Thesis Catalyst Synthesis, KinCat Group, NTNU | Aug 2025 – Dec 2025

Trondheim, Norway | UTFORSK mobility scholarship | Host: Prof. Edd Anders Blekkan

- Prepared the thesis catalyst batches; troubleshot low recovery from filtration losses and an ignition of residual NaBH4 on the filter in a scaled-up batch.
- Revised the route (aqueous NaBH4, less reductant, modified filtration): the third batch gave the highest recovered mass, about 220 mg, with no ignition observed; checked morphology by SEM.

PROJECTS

SNCR Retrofit for NOx Reduction in a Biomass Boiler | Aug 2025 – Nov 2025

NTNU process design project, team of four | Industry partners: Billerud Viken, Vardar Varme

- Carried out the literature survey and built the Python (Cantera) models: equilibrium combustion and plug-flow SNCR kinetics, mapping NO reduction and ammonia slip against temperature and NSR.
- Performed the hand calculations for the design basis and contributed Aspen HYSYS process simulation.
- Idealised modelling predicted above 90% NO reduction; the design adopted a conservative ~60%, keeping modelled NOx within the 300 mg/Nm³ limit.
- The team's design covered reagent selection, equipment sizing, safety measures and CAPEX/OPEX estimates; presented the poster at Treforedlingsforum 2025, Oslo.

DFT Study of the Oxygen Reduction Reaction on a Pt13 Cluster | Jan 2025 – May 2025

IIT Madras | Individual course project | Guide: Dr. Jithin John Varghese

- Optimised O2, OOH*, O*, OH* and H2O on Pt13 in Gaussian 16 (UB3LYP) and confirmed minima by frequency analysis; all step energies were negative in electronic (SCF) energy, without free-energy corrections.

TECHNICAL SKILLS

Electrochemical Testing: three-electrode cells; Autolab potentiostat/galvanostat (NOVA); cyclic voltammetry; chronoamperometry; ECSA; mass activity; durability testing (current retention); Ag/AgCl to RHE conversion

Electrocatalysts and Electrodes: wet-chemical synthesis (aqueous adsorption, NaBH4 reduction); heat treatment under argon; catalyst-ink formulation and Nafion loading; drop-cast electrodes on glassy carbon and carbon paper

Materials Characterisation: FE-SEM, SEM-EDX, XRD (Scherrer size, lattice parameter), N2 physisorption (BET); XPS data analysis (coursework)

Process Engineering: mass and energy balances; technology screening, vendor trials and feasibility assessment; electrocoagulation and electroflocculation; CAPEX and OPEX estimation; process safety

Modelling and Software: Python (Cantera); Aspen HYSYS; DWSIM; CHEMCAD (B.Tech labs); Gaussian 16 and Avogadro (DFT); OriginPro; Excel

Training and Familiarity: HPLC, GC-MS, FTIR and UV-Vis (industrial training); LSV, EIS, chronopotentiometry and TGA (coursework and lab sessions); plant HSE and SOP training

Languages: English (IELTS Academic, overall band 7.0, CEFR C1, July 2026); Hindi

POSITIONS OF RESPONSIBILITY

Deputy Placement and Internship Coordinator | Aug 2024 – Jul 2025

Placement and Internship Cell, IIT Madras

- Handled placements and internships for the 2023–25 and 2024–26 M.Tech batches; 75% of M.Tech chemical engineers were placed.

CONFERENCES AND WORKSHOPS

- Poster on SNCR technology for NOx reduction in a bio-boiler, Treforedlingsforum 2025, Oslo, November 2025.
- Attended EuropaCat 2025, the European Congress on Catalysis, Trondheim, Norway.
- GIAN course: Characterization of Heterogeneous Catalysts, IIT Madras, April–May 2025.
- Paper presentation, SCHEMCON 2021, MANIT and IISER Bhopal, October 2021.

AWARDS AND SCHOLARSHIPS

- GATE 2024, Chemical Engineering: All India Rank 864 of 13,937 candidates.
- UTFORSK mobility scholarship from NTNU for a research semester in Norway, August–December 2025.